Tutorial 2

1. A deck of six cards consists of three black cards numbered $1, 2, 3$, and three red cards numbered $1, 2, 3$. First, Vann draws a card at random and without replacement. Then Paul draws a card at random and without replacement from the remaining cards. Let A be the event that Paul's card has a larger number than Vann's card. Let B be the event that Vann's card has a larger number than Paul's card.

   (a) Are A and B mutually exclusive?

   (b) Are A and B complements of one another?

2. Define a sample space for the experiment of drawing two coins from a purse that contains two quarters, three nickels, one dime, and four pennies. For the same experiment, describe the following events:

   (a) drawing 26 cents;

   (b) drawing more than 9 but less than 25 cents;

   (c) drawing 29 cents.

3. At a certain university, every year eight to 12 professors are granted University Merit Awards. This year among the nominated faculty are Drs. Jones, Smith, and Brown. Let A, B, and C denote the events, respectively, that these professors will be given awards.

   In terms of A, B, and C, find an expression for the event that the award goes to:

   (a) only Dr. Jones;

   (b) at least one of the three;

   (c) none of the three;

   (d) exactly two of them;

   (e) exactly one of them;

   (f) Drs. Jones or Smith but not both.

4. Use Venn diagrams to verify that

   (a) If event A implies B, and B implies C, then A implies C.

   (b) If event A implies B, then B^c implies A^c.

5. A bus arrives at a station every day at a random time between 1:00 P.M. and 1:30 P.M. What is the probability that a person arriving at this station at 1:00 P.M. will have to wait at least 10 minutes?